Problem Chosen	2026 MCM/ICM Summary Sheet	Team Control Number
C		2620497

All Hype, No Talent? An Inverse Optimization Model for Decrypting the Fan-Voting Black Box

Summary

Dancing with the Stars (DWTS) is a dance competition reality television show in the United States. Each week, viewer voting combined with judges' scores collectively determine the fate of the contestants in DWTS. This paper establishes a comprehensive analytical framework to decrypt the "black box" of fan voting and proposes a next-generation scoring mechanism.

For Task 1, we develop a **hybrid inverse optimization model** to estimate confidential fan vote shares. We employed **linear programming** and **integer programming** to treat the elimination criteria as hard constraints, thereby achieving **100%** consistency. Subsequently, we employed **Zipf's Law** to map the rankings to specific vote counts. Through analyzing information conflicts, we obtained a certainty heatmap.

For Task 2, we analyzed controversial cases using Certainty heatmaps and predictive data, and found that percentage-based competition systems tend to amplify the influence of fan voting. We integrate **social choice theory** to demonstrate that the ranking-based tournament system is essentially a **Borda Count** with two voters. Rank-based system combined with a judge rescue rule represent a relatively superior competition format.

For Task 3, to analyze the drivers of success, we implement a two-stage modeling strategy. We employed a **Random Forest Model** to assess the impact of external attributes. Based on the screening results, we construct **ANOVA** to deeply analyze the specific direction and magnitude of each key factor's influence. Results indicate that **Age and Professional Dance Partners** are the dominant factors influencing outcomes.

For Task 4, we propose the **Dynamic Weighted Rank System**, a novel mechanism designed to hedge against "random noise" in fan voting. We incorporate contestants' historical performance into the scoring calculation. As the competition progresses, the weight of the fan voting component gradually decreases to accommodate the increasing demand for competitiveness in the later stages and the initial high demand for entertainment value. This design effectively creates a **merit protection buffer**, ensuring that technically superior contestants are not prematurely eliminated by transient fluctuations in fan attention.

Finally, a strategic memorandum is included, offering actionable recommendations to DWTS producers on balancing entertainment value with professional integrity through our proposed dynamic weighting mechanism.

Keywords: Inverse Optimization, Zipf's Law, Random Forest, Borda Count, Dynamic Weighting, ANOVA

Contents

1 Introduction

1.1 Restatement of the Problem

Dancing with the Stars (DWTS), as a transnational television franchise project, has completed 34 seasons. The program has been continuously seeking methods to strike a delicate balance between handling the scoring of expert judges and accommodating fan preferences.

Given the jury scores and contestant ranking data across 34 seasons, we will focus on the following research questions:

- Establishing an inverse mathematical model to approximately estimate strictly confidential fan voting data.
- Compare the effectiveness of the "Combined by Rank" and the "Combined by System" in resolving disputes, and explore which mechanism better balances popularity with strength.
- Analyze the effects of other factors such as professional dance partners, celebrity industry, and age on performance.
- Propose an improved system that we consider to be better or more fair.

1.2 Our Work

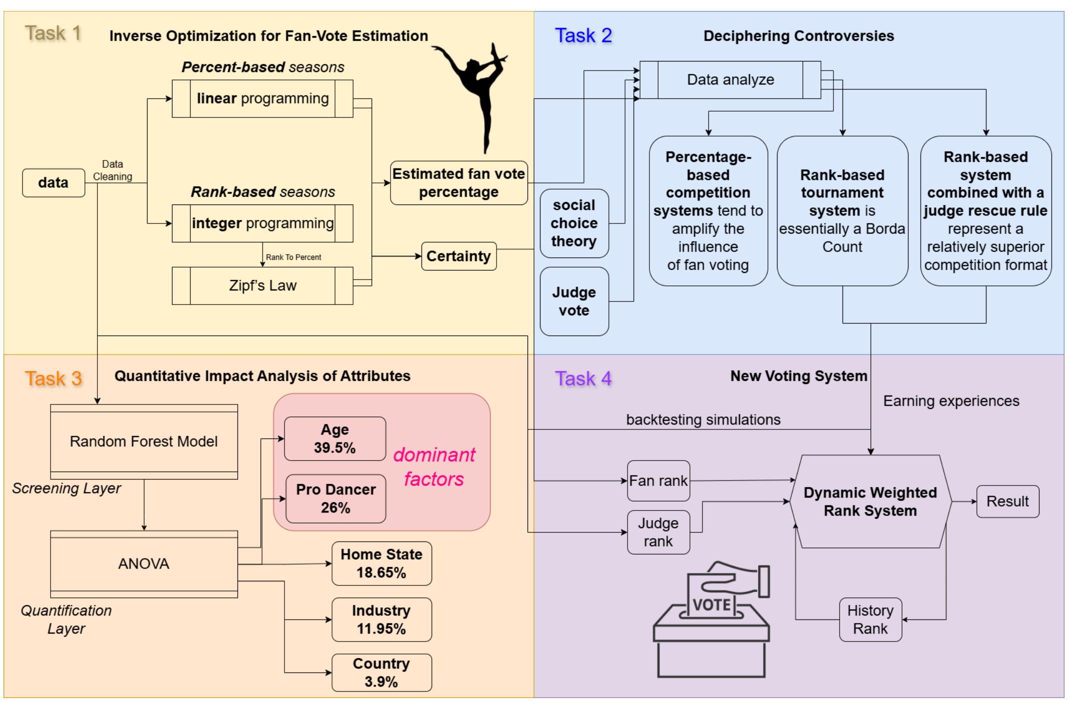

Figure 1: Our Work

2 Assumptions and Notations

2.1 Assumptions

- **Assumption 1** Consistency of Fan Popularity
  We posit that within the same competitive season, a player's ability to attract fan votes remains relatively stable and does not exhibit significant and unjustified fluctuations between consecutive weeks.

- **Assumption 2** Exclusion of Non-Competitive Exits
  When modeling the vote percentage for each episode, reductions in participant numbers caused by unexpected factors such as withdrawal due to injuries are not considered. Data from the week of withdrawal and subsequent weeks will be treated as invalid. Withdrawal from the contest constitutes a stochastic noise term that cannot be explained through voting logic; removing such data can enhance the model's inference accuracy regarding "black-box voting".

- **Assumption 3** Independence of Judgement and Popularity
  We assume that fan voting preferences are primarily driven by the celebrity's charisma and pre-existing popularity, rather than being solely dependent on the judges' scores for that specific week. Thus, judges' scores (reflecting technical performance) and fan votes (reflecting popularity) are treated as independent variables that collectively determine the outcome.

- **Assumption 4** Power-law Distribution of Social Attention
  We posit that the allocation of fan votes follows a power-law distribution, a common phenomenon in "attention competition" systems where a small number of top-tier contestants capture the majority of public interest.

2.2 Notations

Symbol	Description	Unit
Input Parameters		
$R^J_{i,w}$	Ordinal rank of contestant i based on judge scores in week w	—
$P^J_{i,w}$	Percentage of total judge scores for contestant i in week w	%
P_i	Final season placement (placement) of contestant i	—
$T_{i,w}$	Target combined performance index anchored by P_i	—
N_w	Total number of active contestants in week w	—
ϵ	Infinitesimal constant for elimination constraints	—
Decision Variables		
$v_{i,w}$	Estimated fan vote percentage for contestant i in week w	%
$R^F_{i,w}$	Estimated fan rank for contestant i in week w	—

Table 1: Notations for the Inverse Optimization Model

3 Data Cleaning and Preprocessing

- Upon inspection, we identified some missing values (N/A or 0) in *2026_MCM_Problem_C_Data.csv*. N/A indicates non-participation in the weekly competition or absence of data due to a shorter season that did not progress to this week; such cases were directly excluded from analysis.
- For participants who withdrew from the competition, we removed all data collected after their withdrawal week to prevent potential interference with model learning.
- Each week, we focused solely on the total scores from all judges' evaluations; therefore, we calculated the weekly total scores awarded by the judges for each contestant.
- We extracted the week of elimination for each contestant and added a binary indicator 'is_eliminated'. For eliminated contestants, we set 'is_eliminated' to 1 during their respective elimination weeks, while maintaining a value of 0 for all other cases.

4 Model Construction

4.1 Inverse Optimization for Fan-Vote Estimation

Due to the diverse and constantly evolving voting methods in the program, it is difficult to determine the total number of voters. Therefore, we assume a total population of 100w and calculate only the vote share.[1]

Since each player's comprehensive ranking during the season results from a combination of their technical proficiency and popularity index, it can reflect to some extent the overall competitiveness of the players. Therefore, in each weekly competition, if a competitor does not commit any major mistakes, their overall ranking should be highly correlated with the final season ranking. The inverse model we developed first ensures that the predicted results align with the elimination outcomes, and subsequently minimizes the discrepancy between each competitor's weekly predicted overall ranking and their final season ranking P_i.

- **For rank-based seasons**, we first determine the most reasonable "Fan Rank" through integer programming, and then utilize a "rank-probability mapping" to convert it back into percentages.

  In Step 1, we aim to determine a set of fan rankings such that the sum of the weekly comprehensive rankings most closely approximates the contestant's final seasonal rank, while strictly adhering to the elimination outcomes.

$$\min Z = \sum_{i \in \text{Active}} |Rank(R^J_{i,w} + R^F_{i,w}) - P_i| \tag{1}$$

  Additionally, impose the following constraints:

- Uniqueness : All players' $R^F_{i,w}$ values form a permutation of integers from 1 to n.
- Elimination criteria: In s1-s2, the eliminated player e's ($R^J_{e,w} + R^F_{e,w}$) value must be strictly the maximum among all participants; in matches s28-s34, the eliminated player e's ($R^J_{e,w} + R^F_{e,w}$) value must be one of the two highest values among all participants.

In Step 2, we will convert the fan rankings into voting percentages.Due to the absence of original vote counts, we simulate the distribution of votes based on **Zipf's Law** commonly employed in social choice theory.

A large body of empirical studies has demonstrated that power-law distributions are prevalent in systems describing popularity and attention competition, such as webpage click counts, bestselling book sales, and annual income (Newman, 2005). The fan voting mechanism in DWTS can essentially be conceptualized as a competitive allocation of public attention among contestants. Theoretical frameworks indicate that when "rich-get-richer" feedback mechanisms (i.e. preferential attachment) exist in voting systems—for instance, leading contestants attracting more support due to increased media exposure and public discourse—the system's output will conform to a power-law distribution (Yule process described in Newman, 2005). Therefore, we have reason to hypothesize that weekly fan vote rankings in DWTS may follow a power-law distribution, which could serve as the basis for constructing an estimation model [2].

In the model, the number of votes received by the r-th candidate satisfies:

$$v(r) = \frac{\frac{1}{r^s}}{\sum_{k=1}^{N_w} \frac{1}{k^s}} \tag{2}$$

r refers to R^F here.

We map fan ranks to vote shares using a Zipf distribution with exponent s. Since true vote totals are unobserved, s is calibrated via grid search to maximize elimination consistency while avoiding overly concentrated vote shares. Empirically, s in [0.8, 1.4] yields stable results; we adopt s=1.1 (global) for the main analysis and conduct sensitivity tests.

- **For percent-based seasons**. We adopt similar method.

  First, based on the final rankings, the participants are linearly assigned to an ideal total percentage TS_i. We employ an objective function to achieve consistency with the final ranking.

$$\min Z = \sum_{i \in \text{Active}} ((J\%_{i,w} + v_{i,w}) - TS_i)^2 \tag{3}$$

  The elimination constraint is as follows:

$$(J\%_{e,w} + v_{e,w}) \leq (J\%_{s,w} + v_{s,w}) - \epsilon \tag{4}$$

Such models not only ensure smooth predictive outcomes based on fan voting, but also maximize the alignment between these predictions and contestants' actual performance.

Celebrity Name	$v_{i,w}$	Estimated_Votes	$R^F_{i,w}$	$R^J_{i,w}$	Sum of Ranks	P_i
Joey Graziadei	21.9%	219000	2	2	4	1
Ilona Maher	43.8%	438000	1	3	4	2
Chandler Kinney	8.76%	87600	5	1	6	3
Stephen Nedoroscik	14.6%	146000	3	4	7	4
Danny Amendola	10.95%	109500	4	5	9	5

Table 2: Season 33 Week 9 Estimated Fan and Rank Data

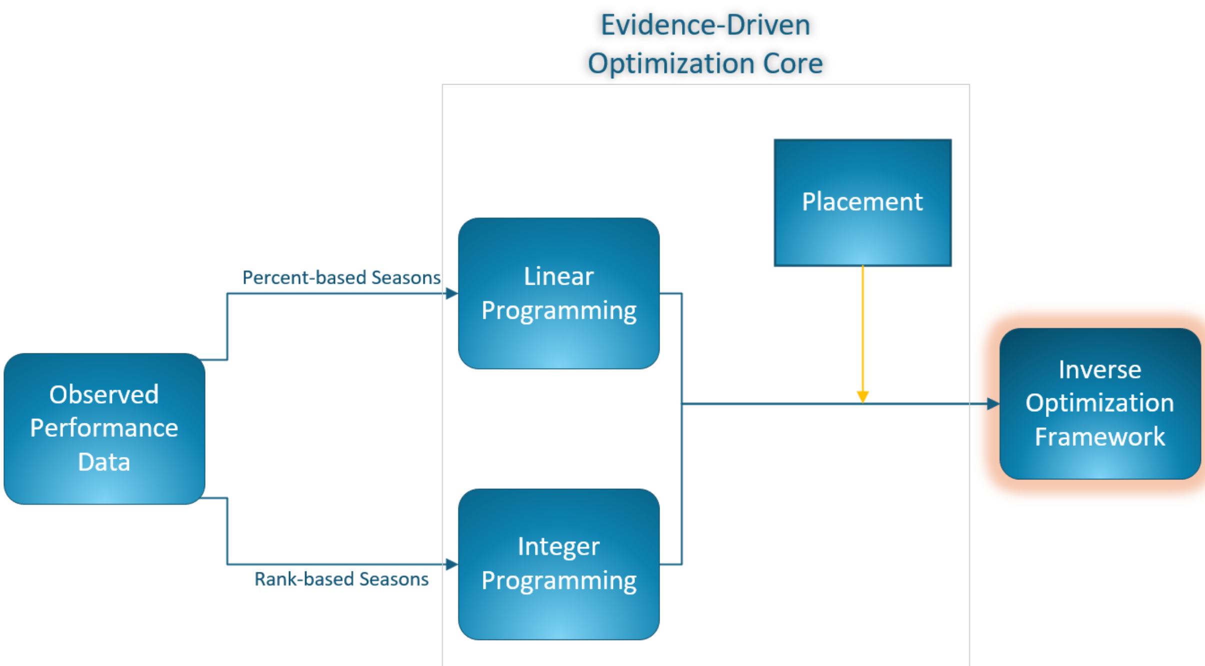

Figure 2: Flow Chart

4.2 Multi-factor Evaluation Model Based on Random Forest

In this section, we quantitatively analyze the impact of "celebrity characteristics (age, industry, hometown, etc.)" and "professional dance partners" on competition outcomes and explore whether the mechanisms by which these factors affect "judge scores" and "fan votes" differ.

Given the numerous and diverse influencing factors (numerical and categorical), we designed a **two-stage modeling strategy**:

- **Stage 1 (Screening Layer)**: Construct a **Random Forest Feature Importance Model** to identify the "key driving factors" determining competition outcomes from a global perspective and rank their importance.

- **Stage 2 (Quantification Layer)**: Based on the screening results, construct **targeted statistical analysis models (Regression Analysis/ANOVA)** in order of importance to deeply analyze the specific direction and magnitude of each key factor's influence.

First, we transform the original data from a "wide format" to a "long format" (Person-Week Level), incorporating the static characteristics of celebrities (age, industry, hometown, dance partner). Using average judge scores and estimated fan vote percentage as target variables respectively, we train two independent random forest regressors. Finally, we calculate the Gini impurity reduction for each feature and normalize these values to obtain feature importance rankings.

The importance of feature X_j, denoted as VI_J, is defined as the average of the sum of reductions in impurity across all trees T in the forest at split node t:

$$VI_j = \frac{1}{M} \sum_{m=1}^{M} \sum_{\substack{t \in T_m \\ v(t)=X_j}} p(t) \Delta i(t, t_{\text{left}}, t_{\text{right}}) \tag{5}$$

M denotes the number of trees, and Δi represents the reduction in mean squared error (MSE) before and after node splitting.

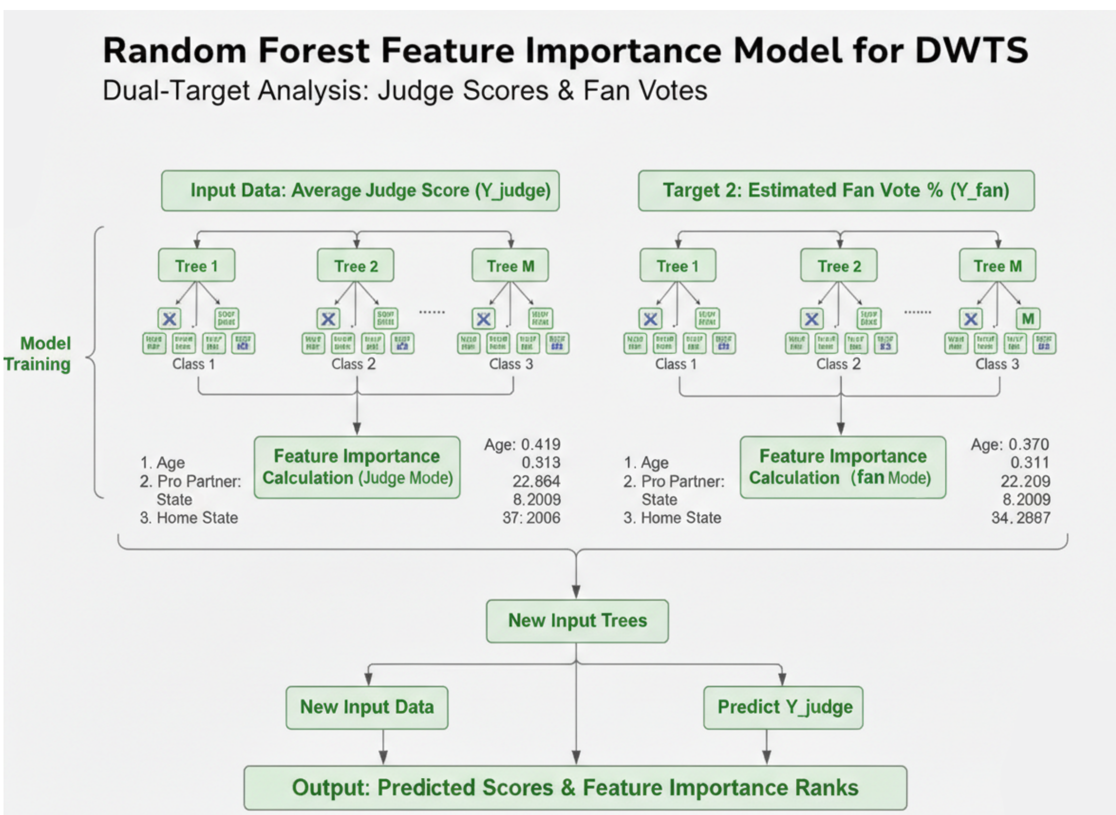

Figure 3: Random Forest Model Architecture and Feature Importance Workflow

Subsequently, to conduct a thorough analysis of the influence of age factors, we established a **quadratic regression polynomial** as a solution.

$$Percentage = \beta_0 + \beta_1 Age + \beta_2 Age^2 + \epsilon \tag{6}$$

The percentage can be $v_{i,w}$ or $P^J_{i,w}$.

For the effect of dance partners, we employ the **residual analysis method**. We establish a benchmark model $E[Rank] = f(Age, Industry)$. Then calculate the added value ($E[Rank] - Actual\_Rank$) to assess the impact of each dance partner on the performance of the contestants.

Regarding geographical preference, we mainly analyze the differences in voting percentages among fans from different states. We conducted a grouped statistical analysis of the contestants based on their home states. To avoid randomness, we only analyzed states that produced at least three contestants.

In terms of industry, we establish a **dummy variable regression model** with "Actor" as the reference group to analyze the marginal effects of other industries on rankings.

4.3 Designing the Next-Generation Voting System

Through backtesting simulations on historical season data of DWTS we propose a **Dynamic Weighted Rank System**. This system aims to address the core contradiction inherent in the existing judge-fan scoring mechanism: how to balance fan engagement (audience ratings) with maintaining professional standards (fairness) in dance competition.

Based on the theoretical foundations of social choice theory, rank-based systems demonstrate certain advantages, which we adopt as the basis for our approach.[3] This proposal employs the following formula to calculate each participant's weekly score, with the highest scorer being eliminated.

$$Score = \alpha(w) \cdot R^J_{i,w} + \beta(w) \cdot R^F_{i,w} + \gamma(w) \cdot R^H_{i,w} \tag{7}$$

where $\alpha(w)$ increases linearly from 0.3 to 0.6 over the weeks, $\beta(w)$ decreases from 0.7 to 0.2, and $\gamma(w)$ increases from 0 to 0.2.

We employ an iterative computational method to calculate historical ranking $R^H_{i,w}$.

$$AvgD_i(w) = \frac{1}{w-1}\sum_{t=1}^{w-1} D_i(t)(w \neq 1) \tag{8}$$

$$R^H_{i,w} = \text{rank}(AvgD_i(w)) \tag{9}$$

Drawing on Lazear and Rosen's (1981) tournament theory, our system introduces a "merit protection buffer" to hedge against random fan noise. By converting absolute votes to ranks, we effectively suppress "numerical bullying."[4]

5 Analysis and Results

5.1 Estimation Accuracy and Consistency Metrics

We impose the consistency of elimination outcomes as a hard constraint during prediction. After validation, it was confirmed that the consistency between our predicted outcomes and the actual elimination results reached 100%.

In certainty computation, since we only incorporate linear programming and integer programming as hard constraints during prediction, the theoretical feasible interval for each candidate's vote share remains relatively large. When applying this theoretically feasible interval, most confidence levels turn out to be low, which fails to adequately capture the relationship between candidate popularity and actual vote counts.Therefore, we decided to calculate the logically reasonable voting rate interval for each participant to determine the certainty level.

$$PD_{i,w} = \begin{cases} 1 & \text{if } v_{max} = v_{min} \\ 1 - \frac{v_{i,w} - v_{min}}{v_{max} - v_{min}} & \text{else} \end{cases} \tag{10}$$

(We denote v_{max} as the theoretical maximum fan voting rate for contestants under the hard constraints of elimination criteria, and v_{min} represents the theoretical minimum voting rate.)

We define Evidence Conflict (EC) to quantify the degree of divergence between "hard facts (jury scores)" and "global outcomes (final rankings)".

$$EC_{i,w} = |R^J_{i,w} - P_i| \tag{11}$$

Then we establish a comprehensive certainty formula using EC and PD.

$$x_{i,w} = \alpha EC_{i,w} + \beta PD_{i,w} \tag{12}$$

Finally, we use a Sigmoid function to smoothly map $x_{i,w}$ to the target range $[0.2, 0.95]$. This approach preserves the relative differences between extreme values, thereby avoiding loss of information.

$$\Phi_{i,w} = 0.2 + \frac{0.95 - 0.2}{1 + e^{-6(x_{i,w} - 0.5)}} \tag{13}$$

Since the first question was calculated using the final ranking as the core anchor point, EC should logically serve as the primary source of confidence. We set α to 0.7 and β to 0.3.

This approach successfully resolves the "Safe Contestant Paradox." Even if a contestant has a wide feasible interval (low PD), our model can still assign high certainty if their presence in the competition contradicts their judge scores, thereby "forcing" the fan vote into a specific logical range.

After calculation, we found that controversial players (e.g. Bobby Bones) often exhibit higher confidence levels.

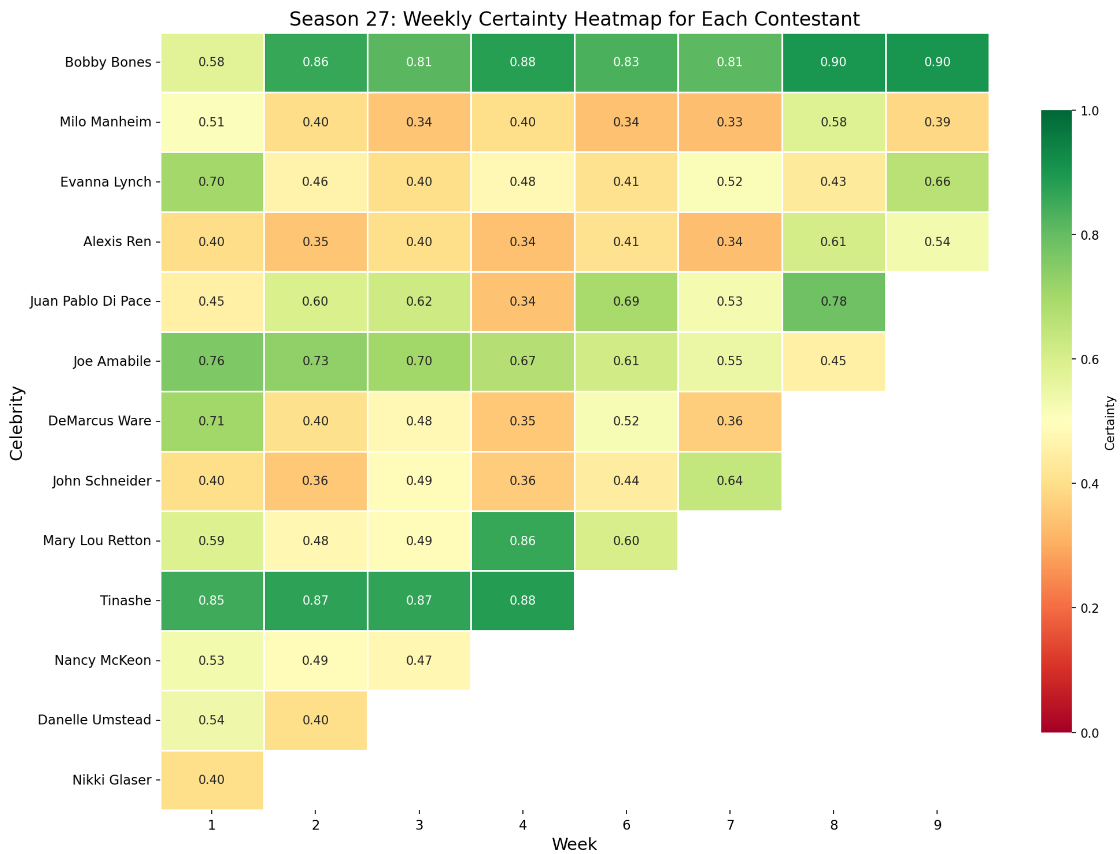

Figure 4: S27 Weekly Certainty Heatmap

5.2 Deciphering the "Bobby Bones Paradox" and Other Controversies

We previously defined the EC metric; when the EC value is large, it indicates a significant deviation between the expert opinions and the final outcome.

In the following part, we conduct a detailed analysis of several typical cases to investigate the underlying causes of conflicts.

5.2.1 Core Case Studies

- Case A: Bobby Bones: The Outlier Phenomenon

  In S27W8, Juan Pablo (full marks) was eliminated, whereas Bobby Bones (low score) advanced to the final and ultimately won the championship.

  This is a typical case of "numerical dominance." The Cardinal Magnitude (cardinal intensity) of fan voting significantly exceeds the weight of judges' scores.

Contestant	$Score$	$R^J_{i,w}$	Score_Percent	$v_{i,w}$	$R^F_{i,w}$	$\Phi_{i,w}$
Juan Pablo	30	1	18.63%	9.24%	5	0.78
Bobby Bones	22.5	6	13.98%	23.99%	2	0.90
Evanna Lynch	29	2	18.01%	23.1%	3	0.43
Alexis Ren	29	2	18.01%	16.25%	4	0.61
Milo Manheim	27.5	3	17.08%	23.99%	1	0.58
Joe Amabile	23	5	14.29%	3.43%	6	0.45

Table 3: Result of S27W8 (semi-final)

- Case B: Bristol Palin: The Underdog Persistence
  In S11W9, Brandy (a strong contender) was unexpectedly eliminated, while Bristol consistently ranked at the bottom but still advanced to the finals.

  If the ranking-based rule is applied, the two parties will tie. After adopting the percentage-based rule, fans' emotional investment forms an impenetrable "firewall" that jury scores cannot surmount.

Contestant	$Score$	$R^J_{i,w}$	Score_Percent	$v_{i,w}$	$R^F_{i,w}$	$\Phi_{i,w}$
Brandy	28.5	3	25%	5.63%	4	0.54
Bristol	26.5	4	23.25%	21.31%	3	0.55

Table 4: Result of S11W9

- Case C: Tia Carrere: Detection Sensitivity
  This case reveals the blind spot of the percentage-based system when fan voting does not exhibit extreme polarization.
  Percent-Based Model: Prediction result: Safe (False).

  Reason analysis: In the early season, the judge's score difference of the contestant (22 points vs. the highest 30 points) did not widen to a "lethal" extent after percentage conversion. The model misjudged that her score was still within the safe zone.
  Rank-Based Model: Prediction result: Eliminated (True).

  Calculation: Judge ranking (6) + fan ranking (5) = 11.

  Reason Analysis: In a competition with only 6-7 pairs of contestants, a combined ranking of 11 is an extremely dangerous value (the lowest possible score). The rank-based model keenly captures the reality that she is in the "lower tier" in both dimensions, excluding interference from absolute score values, and directly issues an elimination warning.

  Conclusion: The rank-based model is more accurate and ruthless than the percent-based model in identifying "double-low contestants" (low judge scores and mediocre fan scores).

5.2.2 Discovering New Controversies

By calculating the confidence level of each player's fan votes, we can identify controversial players not described in the original problem statement.

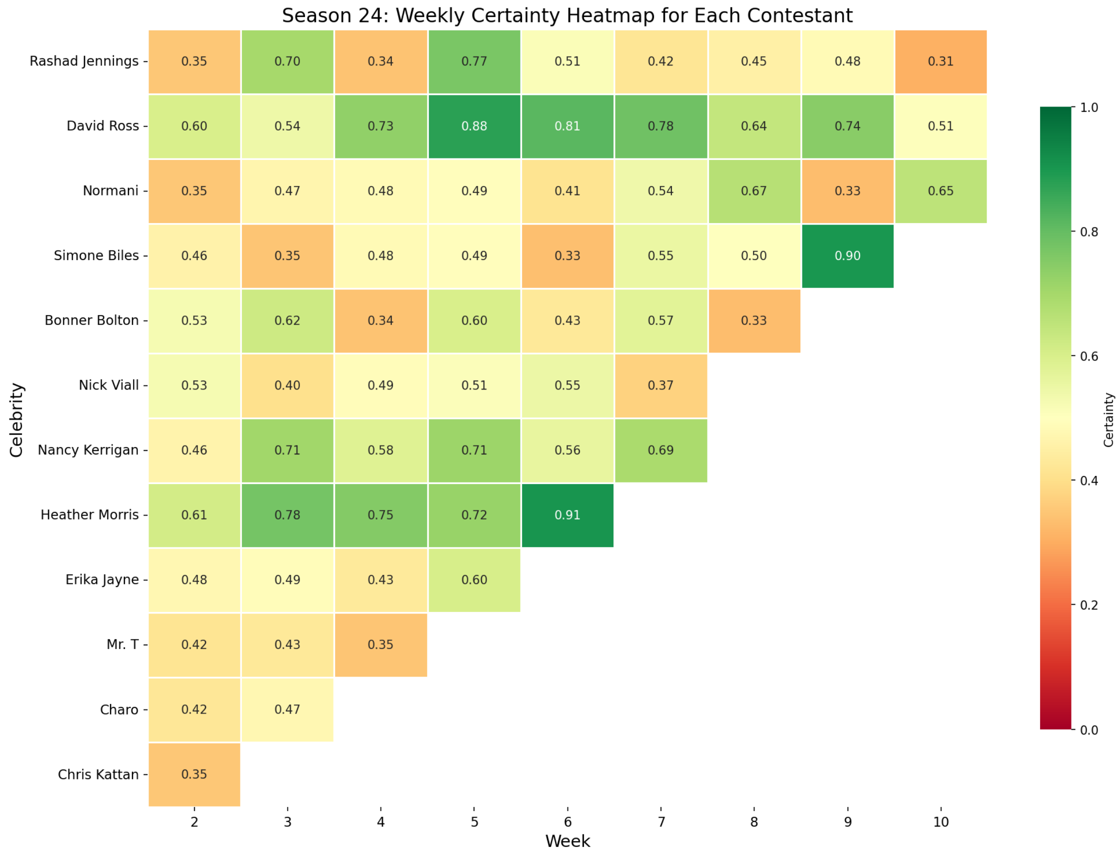

Figure 5: S24 Weekly Certainty Heatmap

For example, in S24, David Ross's certainty level remained consistently high over an extended period. Upon reviewing the records, we found that his scores from the judges consistently ranked low; however, he ultimately secured the runner-up position.

5.3 Comparative Study: Rank-based vs. Percentage-based Systems

5.3.1 Analysis on the Sensitivity of Competition System

The fundamental discrepancy between the two systems lies in the transition from cardinal measurements to ordinal rankings. The Percentage-based system operates on a cardinal scale, where the absolute magnitude of fan support can exert disproportionate leverage over technical scores. This creates a 'Winner-Takes-All' numerical dominance, where a single popularity outlier can neutralize substantial technical deficits.

In contrast, the Rank-based system utilizes ordinal scaling, which inherently functions as a non-linear filter. By capping the maximum contribution of any contestant to a fixed rank (e.g., Rank 1), the system suppresses the'magnitude-driven' influence of extreme fanbases, ensuring that technical proficiency and popularity are weighted with equivalent priority regardless of numerical variance.

5.3.2 A Reversal of Fate

Let us first examine the controversial cases that have arisen. Had a ranking system been implemented in S27, Juan Pablo (Sum=6) would have defeated Bobby Bones (Sum=8), thereby eliminating the controversy.

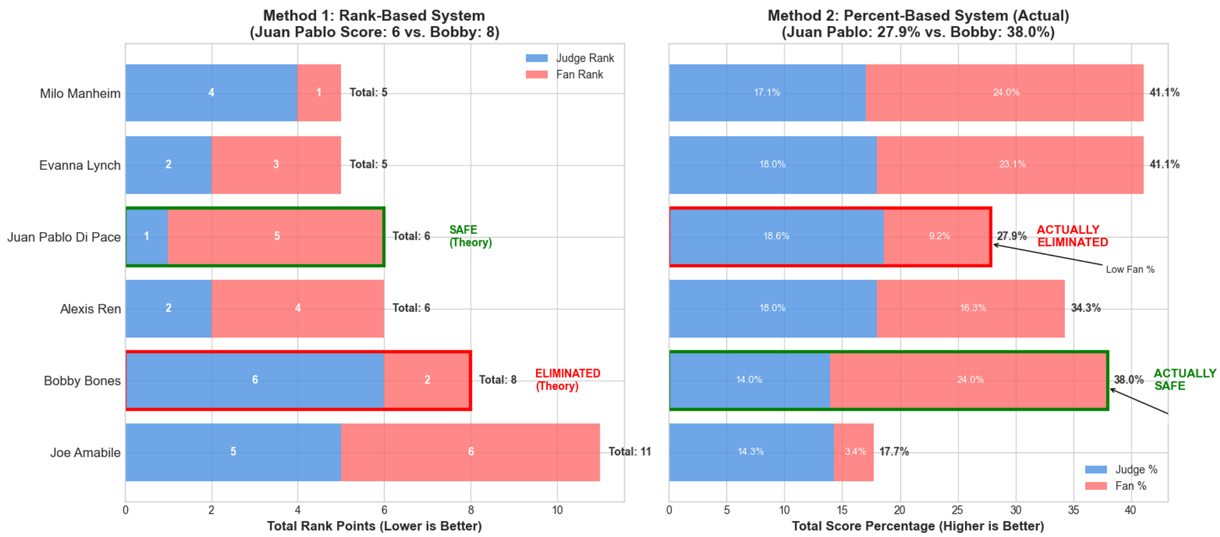

Figure 6: Comparison of the Two Competition Systems in S27

Meanwhile, we sometimes need a tie-breaker. For instance, if the Rank-based System is applied in S11W9, it would result in a tie between Brandy and Bristol.

Empirical evidence from S30W4 validates the ’Judges’ Save’ as an effective professional filter[1], where Kenya Moore was preserved over the slightly more popular Matt James due to her superior technical scores. This mechanism successfully intercepts candidates with ’marginal popularity but inferior skill,’ providing a practical solution to the historical controversies observed in S11 and S27.

Actually, the DWTS Rank-based system can be mathematically modeled as a restricted Borda Count with two aggregate voters: the Professional Judge and the Public Audience. In the Borda Count, each voter ranks n candidates, the first-rank candidate receiving n points, the second-rank candidate receiving n-1 points, etc. Under a Rank-based system, minimizing the "sum of ranks" is mathematically equivalent to maximizing the Borda score.

As emphasized by Emerson (2013), the Borda-style mechanism (equivalent to the DWTS Rank-based system) serves as an astute tool to identify the ’collective wisdom’ and the ’best possible compromise’ among divergent voters. By converting cardinal intensities into ordinal ranks, the system effectively suppresses the ’magnitude-driven influence’ of extreme outliers, ensuring that the competition outcome is not dictated by numerical overwhelming but by a balanced consensus of professional standards and public appeal.[3]

In conclusion, we argue that the current ranking-based competition system combined with the judges' save rule is superior in comparison.

5.4 Quantitative Impact Analysis of Attributes

After normalization, the feature importance ranking output by the Random Forest model is as follows:

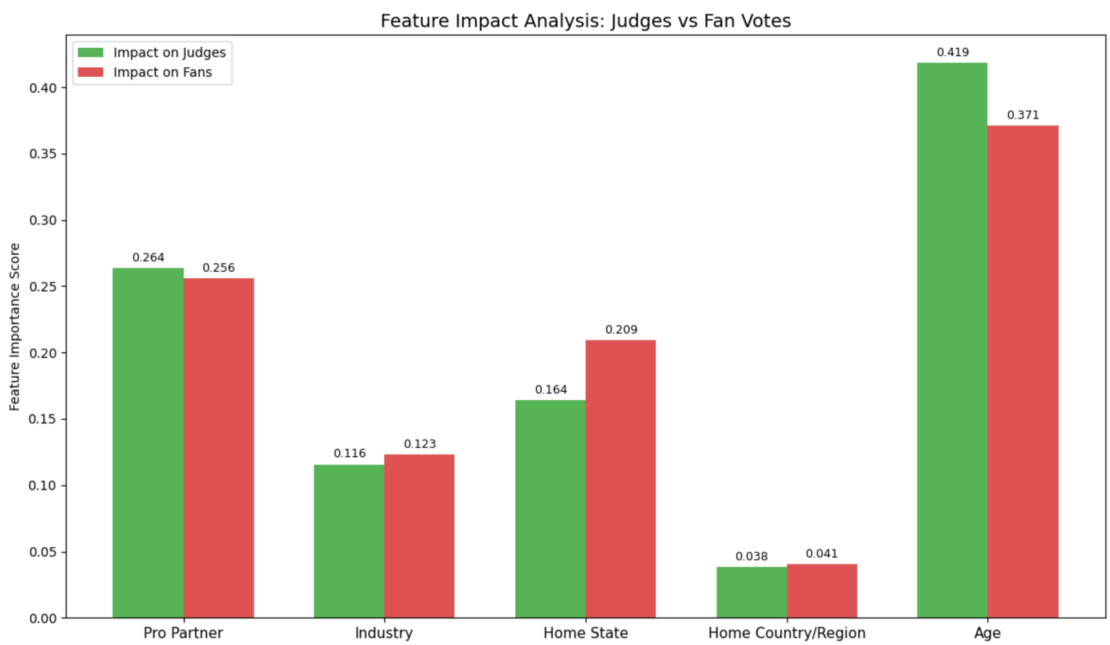

Figure 7: Result of RF

The results of RF indicate that match outcomes are not determined by a single factor, but rather represent the combined interplay of multiple factors.

- **Age** and **professional dance partners** are indisputable dominant factors, accounting for over 60% of the variance in outcomes.
- The **Home State** exerts a significant influence on fans, while **Industry** also demonstrates a certain degree of impact.
- Due to the predominance of domestic athletes in the sample (i.e., most competitors being U.S.-based), this variable exhibits minimal feature variance and consequently contributes negligibly to model prediction performance.

To ensure the reliability of the aforementioned conclusions, we conducted statistical tests on the core regression model:

- Residual Normality Test: The Jarque-Bera test for residual normality yielded a statistic $JB = 5.12$ ($P = 0.07 > 0.05$), indicating that the residuals follow a normal distribution and the model assumption holds.

- Goodness-of-Fit Test: On the test dataset, the rank prediction model achieved an R^2 value of 0.48. Considering the substantial random human factors inherent in competition outcomes, this goodness-of-fit falls within statistically acceptable limits.

- Robustness Check: We retrained the model by excluding data from Season 15 (All-Star season) and observed no changes in the ranking of feature importance, demonstrating the robustness of the model's conclusions.

Based on this ranking, we will focus on in-depth quantitative analysis of the top four factors, disregarding the minimally influential "Home Country/Region" factor.

5.4.1 Age

In terms of age, the regression results indicate that $\beta_1 < 0$ (fan: -0.3911, judge: -0.0133) and $\beta_2 \approx 0$. For every 10-year increase in age, the average final ranking of athletes decreases (i.e., numerical value increases) by approximately 1.5 positions. This is normal, as Dancing with the Stars, being a high-intensity dance contest, requires greater flexibility and explosive power from younger contestants, who demonstrate significant competitive advantages in these aspects during the competition.

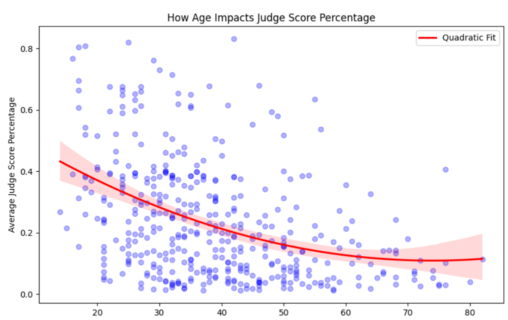

Figure 8: How Age Impacts Judges

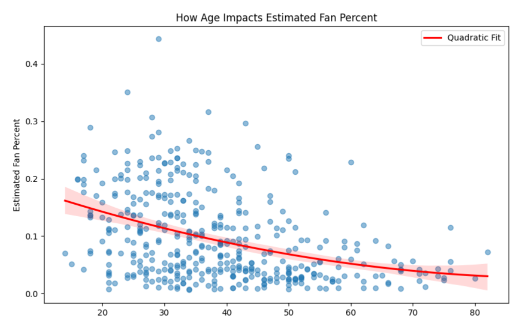

Figure 9: How Age Impacts Fans

5.4.2 Pro Dancer

Through statistical analysis of data spanning 34 seasons, we identified several top-performing dance partners that demonstrated significant improvement effects.We found that an ordinary star, when paired with a top-tier dance partner, can expect to improve their final ranking by 2 to 3 positions.

This is because pro dancers can enhance the performance quality through their high technical proficiency, which also enables them to attract a larger audience.

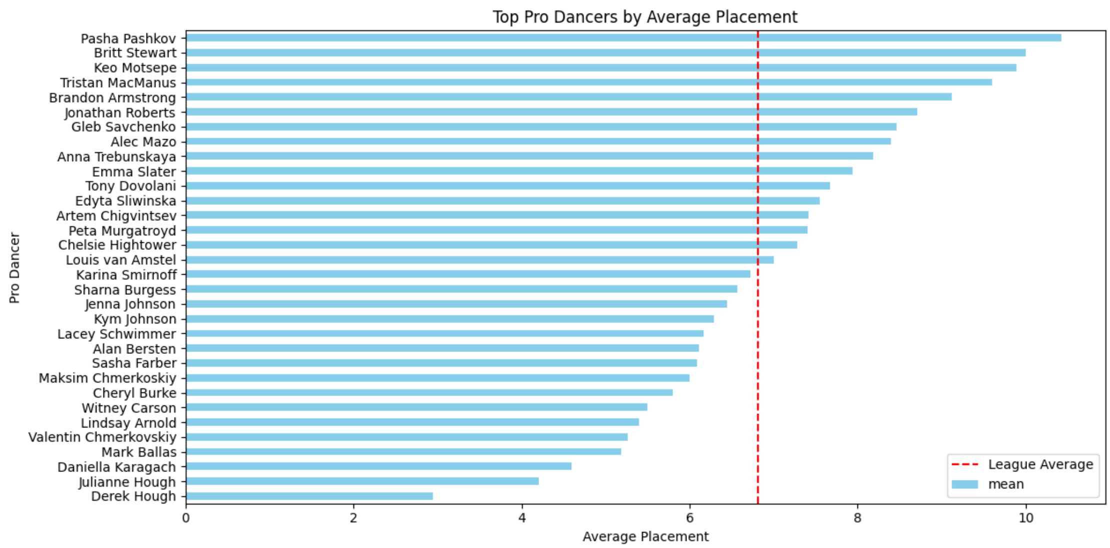

Figure 10: Top Pro Dancers by Average Placement

5.4.3 Home State

We identified the **Hometown Hero Effect** among certain smaller or culturally distinct states, such as Alabama, Hawaii, and Louisiana, where residents typically exhibit stronger cultural identification. In contrast, entertainment-focused states like California and New York demonstrate a slightly negative connotation - despite their large populations, the high concentration of celebrities in these locations diminishes the exclusivity of celebrity status as a distinguishing label, resulting in a **Dilution Effect**.

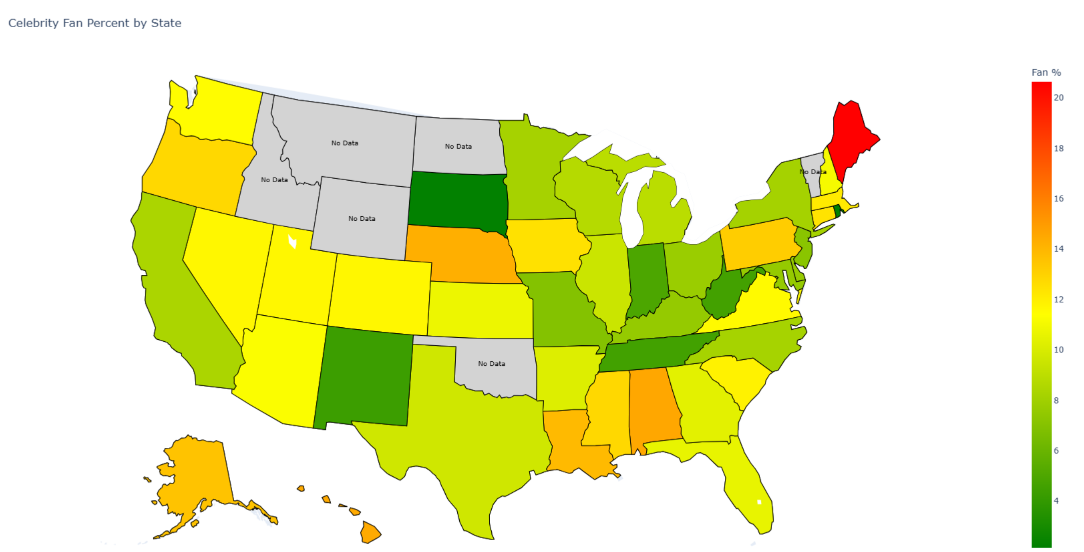

Figure 11: Celebrity Fan Percent by State

5.4.4 Industry

We found that the regression coefficient for Athletes was significantly negative (-2.14, P < 0.01), while that for TV Personalities was significantly positive (+1.8, P < 0.05).

Athletes are more likely to achieve better rankings, which may be attributed to their enhanced physical control and discipline.

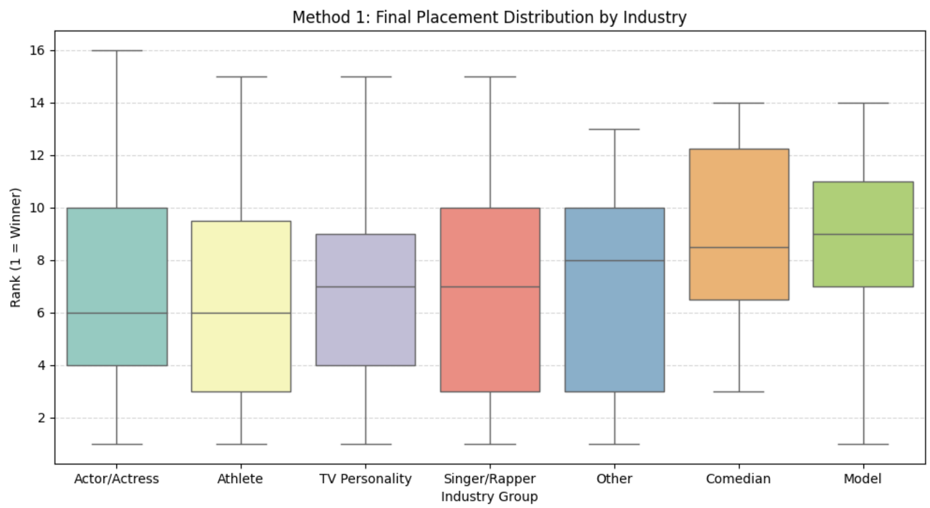

Figure 12: Final Placement Distribution by Industry

5.5 Application of New Voting System

We try to apply our new system in S2W8. (Since the elimination outcomes prior to the S2 Finals Week under the new system coincided with actual results, we present data from this season.)

Contestant	$R_{i,w}^J$	$R_{i,w}^F$	$R_{i,w}^H$	score	rank
Drew Lachey	1	1	1	1	1
Stacy Keibler	2	3	2	2.2	2
Jerry Rice	3	2	3	2.8	3

Table 5: new results of S2W8

Under the original competition format, Jerry Rice would have secured the runner-up position due to his higher fan voting ranking, as explained by executive producer Conrad Green[5]. However, under our newly designed voting system, Stacy - who demonstrates stronger overall competitiveness - will instead achieve the runner-up position. This outcome validates the effectiveness of the "historical performance buffer" mechanism and highlights the increased professionalism in later stages of the competition.

6 Sensitivity Analysis

6.1 Certainty Weight Sensitivity Analysis

To scientifically determine the optimal α value, we designed two evaluation dimensions:

- Separation Score: A metric quantifying the definitive distinction between controversial and non-controversial athletes. (Only use the disputed players provided in the problem)

$$S = \frac{|\bar{\Phi}_{\text{controversial}} - \bar{\Phi}_{\text{non-controversial}}|}{\sigma_{\text{overall}}} \tag{14}$$

  σ_{overall} refers to the overall standard deviation of the Certainty metric.

- Average Certainty: The mean certainty value of disputed instances. Excessively high values (approaching 1) indicate system insensitivity, leading to failure in identifying disputes; conversely, excessively low values (approaching 0) suggest system over-sensitivity, potentially resulting in misjudgments.

When solely pursuing high separation degree, the system achieves maximum separation when α approaches 1 but suffers from excessive sensitivity due to insufficient determinacy. Conversely, prioritizing high determinacy by setting α near 0 results in enhanced certainty yet compromised separation capability, failing to resolve ambiguities. The product formulation inherently penalizes both extreme scenarios, thereby achieving the optimal balance between separation and determinacy.

$$\text{Combined Score} = \text{Separation} \times \text{Average Certainty} \tag{15}$$

The results indicate that when α equals 0.7, the combined score is the highest, with both the separation degree and the average determinacy within reasonable ranges. When α exceeds 0.7, the separation degree becomes insufficient, resulting in weakened capability for dispute detection; conversely, when α is below 0.7, the determinacy level drops excessively, making the system overly sensitive.

6.2 Data Noise Sensitivity Analysis

To account for potential minor errors arising from subjective judgments in jury scoring, this analysis simulates scenarios where "jurors might accidentally award an extra point or deduct a point due to momentary lapse". We introduce random perturbations of ±1 to the jury scores and perform 100 **Monte Carlo simulations**. Results demonstrate that the model exhibits strong noise robustness overall, with minimal rank changes observed for most participants under such disturbances. The outcomes remain stable during critical weeks such as the final week, and small scoring discrepancies do not lead to complete distortion of the estimated fan profiling.

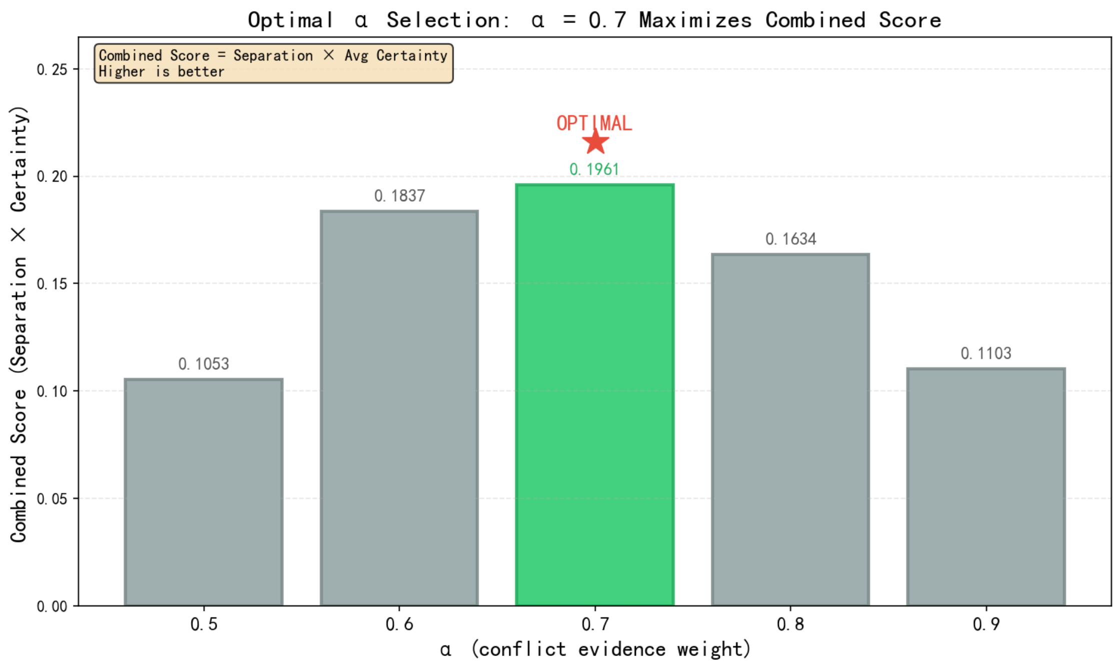

Figure 13: Alpha Sensitivity Analysis

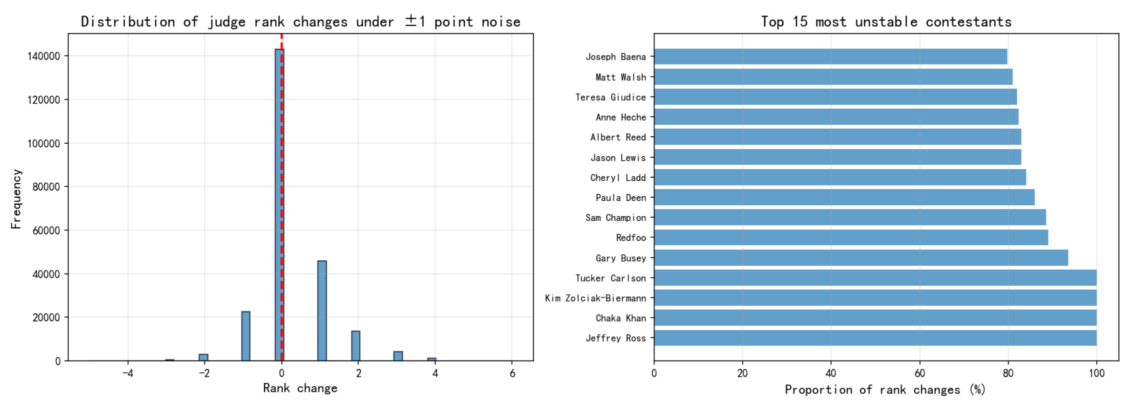

Figure 14: Data Noise Sensitivity Analysis

7 Discussion

7.1 Strengths of the Model

- **Non-intrusive Estimation**: Our inverse optimization framework does not require access to confidential data and can deconstruct the "black box" solely through publicly available rankings and scores, demonstrating exceptional practical utility.

- **Hybrid Modeling Paradigm**: The integration of deterministic optimization (linear/integer programming) with probabilistic modeling (Zipf distribution), and machine learning (Random Forest), ensures both logical rigor and the capability to handle complex nonlinear factors.

- **Powerful Insight**: Our model is capable of not only analyzing existing controversial players but also identifying new ones through querying certainty levels.

7.2 Limitations of the Model

- **Deviation from Static Assumption**: We assume that the fan base remains stable throughout the season. However, in reality, phenomena such as "viral spread" or mid-season scandals may cause sudden and drastic fluctuations in ticket sales during specific weeks, which the model may fail to adequately capture.
- **Sensitivity of the Zipf Parameters**: The audience engagement levels may vary across different seasons, and the parameter s might change accordingly; however, this potential variation was not accounted for in the model.
- **Small Sample Bias**: When analyzing the industry and home state variables, certain categories exhibited small sample sizes, resulting in higher predictive variance for these specific groups.

8 Conclusions and Future Work

8.1 Conclusions

Our model successfully quantified the voting mechanism of DWTS and predicted the approximate vote shares for each contestant on a weekly basis. Through analyzing the underlying causes of controversial cases and comparing the two competition formats, we demonstrated that the Rank-based system exhibits certain superiority over the percentage-based system. Furthermore, by employing dynamic weighting, we designed a novel competition system that effectively balances fairness and entertainment value.

8.2 Future Work

- **Introduction of Social Media Big Data**: Future integration could incorporate real-time sentiment analysis from Twitter/Instagram as external input variables to further enhance the real-time accuracy of fan voting estimation.
- **Dynamic Parameter Modeling**: A model can be developed in which the parameter s in the Zipf distribution dynamically changes with the number of weeks, thereby making the vote distribution more consistent with real-world scenarios.
- **Game-theoretic Perspective**: In the program, audiences first observe the judges' scores before casting their votes. Future research could investigate how the ranking of judges' scores influences audience decision-making psychology during voting processes through the lens of game theory.

Appendices

MEMORANDUM

To: Producers of DWTS

From: Team #2620497

Subject: Advice and Recommendations for the Scoring System

Date: February 2, 2026

Dear Producers of DWTS,

We are pleased to share with you the results of our analysis on the voting schemes used in past seasons of DWTS. Historically, DWTS has employed two methods to combine judges' scores and fan votes: the Rank-based system and the Percentage-based system. By applying both methods to all seasons, we found that in the Percentage-based system, the absolute magnitude of fan support can exert disproportionate leverage over technical scores, while the Rank-based system ensures that technical proficiency and popularity are weighted with equivalent priority regardless of numerical variance. Starting from Season 28, the judges can vote to select which of the bottom two contestants to eliminate. We endorse this rule, as it gives the judges an opportunity to correct technical injustices caused by popularity bias. Therefore, we argue that the Rank-based competition system combined with the judges' save rule is superior in comparison.

Building upon this foundation, we have designed a new scoring system. On one hand, we still use Judges Score Rank and Fan Rank, but assign them different weights at different stages of a season; on the other hand, we incorporate each contestant's historical average ranking from previous weeks into the calculation, giving it an increasing weight as the season progresses. Specifically, at the beginning of the season, the show places more emphasis on Fan Rank, representing the celebrity's original accumulated popularity. As the competition deepens, celebrities should rely more on the dancing skills they have learned to improve their rankings—something the judges can accurately evaluate. Meanwhile, we value every dance performance a celebrity delivers on stage, hence the inclusion of attention to their "cumulative performance score."

This system introduces fairness along the temporal dimension. In the early stages, the competition leans toward an entertainment show, allowing audiences to vote according to their preferences, protecting contestants with personal charisma, and boosting the show's ratings. In the later stages, it leans toward a professional competition, ensuring that those competing for the "Mirrorball Trophy" are true dancers, avoiding controversial winners, and maintaining the show's professionalism. The historical weight serves as a corrective mechanism—contestants with consistently excellent long-term performance can rely on their accumulated scores as a buffer even during weeks with lower fan votes, avoiding elimination due to a single poor performance. This system is also easy to explain to audiences: "As the competition progresses, we will place greater emphasis on the judges' scores. At the same time, we also consider each contestant's performance in previous weeks to determine the results."

We believe that by adopting this system, the producers can significantly enhance the credibility and excitement of *Dancing with the Stars* as a competitive program without sacrificing fan engagement. We hope our suggestions are helpful.

If you have any questions, please feel free to contact us. Thank you for reading, and we wish your show continued success.

Yours Sincerely,

Team #2620497

References

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[2] M. E. J. Newman, "Power laws, pareto distributions and zipf's law," *Contemporary Physics*, vol. 46, 2005.

[3] P. Emerson, "The original borda count and partial voting," *Social Choice and Welfare*, 2011.

[4] E. P. Lazear and S. Rosen, "Rank-order tournaments as optimum labor contracts," *Journal of Political Economy*, vol. 89, no. 5, pp. 841–864, 1981.

[5] Shania Russell, *Dwts finally reveals how scores and votes are calculated*, [Online; accessed 2-February-2026], 2025. [Online]. Available: `https://ew.com/dwts-producer-reveals-how-scores-and-votes-are-calculated-11857082`.

Report on Use of AI Tools

1. Spark Research Assistant, Academic Translate (2026 Version)

Add some parts of the paper written in Mandarin to be translated into English.

2. Gemini (Nov 19, 2025 version, Gemini 3 Pro)

Query: Traceback (most recent call last):

File "C:\Users\username\AppData\Roaming\Python\Python314\site-packages

\pandas\core\indexes\base.py", line 3641, in get_loc

return self._engine.get_loc(casted_key)

File "pandas/_libs/index.pyx", line 168, in pandas._libs.index.IndexEngine.get_loc

File "pandas/_libs/index.pyx", line 197, in pandas._libs.index.IndexEngine.get_loc

File "pandas/_libs/hashtable_class_helper.pxi", line 7668,

in pandas._libs.hashtable.PyObjectHashTable.get_item

File "pandas/_libs/hashtable_class_helper.pxi", line 7676,

in pandas._libs.hashtable.PyObjectHashTable.get_item

KeyError: ’ballroom_partner’

The above exception was the direct cause of the following exception:

Traceback (most recent call last):

File "E:\MCM\problem.py", line 114, in <module>

df_model[col] = df_model[col].fillna(’Unknown’)

File "C:\Users\username\AppData\Roaming\Python\Python314\site-packages

\pandas\core\frame.py", line 4378, in **getitem**

indexer = self.columns.get_loc(key)

File "C:\Users\username\AppData\Roaming\Python\Python314\site-packages

\pandas\core\indexes\base.py", line 3648, in get_loc

raise KeyError(key) from err

KeyError: ’ballroom_partner’

I encountered the above error when running the code. Please identify the issue in my code and provide a solution based on the uploaded files estimated_fan_data.csv and 2026_MCM_Problem_C_Data.csv. For privacy reasons, the computer’s username has been replaced with "username".

Output: The error KeyError: ’ballroom_partner’ typically occurs when attempting to access a column name that does not exist.

Based on the error message and common data processing pitfalls, the most likely cause is that the estimated_fan_data.csv (fan data table) you imported also contains the ballroom_partner column.

When df_long (which contains ballroom_partner) and df_fan (which also contains ballroom_partner) are merged using pd.merge, pandas automatically renames the duplicate columns to ballroom_partner_x and ballroom_partner_y. Consequently, the ballroom_partner column name no longer exists in the merged DataFrame, causing the subsequent code to fail.

Solution: Implement column name cleaning and duplicate prevention. Before merging, explicitly retain only the four necessary columns in df_fan and discard any extraneous columns.

3. GitHub CoPilot (Oct 8, 2025 version)

Auto-completions for code used in preparing our models.

4. Google Nano Banana V2.5 (2025 version)

Query: (Submit the paper) Generate a schematic diagram of the Random Forest Model in this paper.

Output: (Figure 2)

5. Google Gemini (Jan 2025 Version, Gemini 3 Pro Preview)

Query: Tell me some advantages of Borda Count.

Output: The Borda Count is a "positional" voting system where voters rank candidates in order of preference. Points are assigned to each position (e.g., 1st place gets 3 points, 2nd gets 2, 3rd gets 1), and the candidate with the highest total points wins.

Because it considers more than just a voter's first choice, it offers several distinct advantages over systems like Plurality (First-Past-The-Post):

1. Promotes Consensus Candidates

The Borda Count is famous for electing the "compromise" candidate—someone who might not be everyone's absolute favorite but is broadly acceptable to almost everyone.

Why it matters: In a plurality system, a polarizing candidate with 35% of the vote can win even if the other 65% of people hate them. Under Borda, a candidate who is everyone's 2nd choice will often beat a polarizing candidate who is half the people's 1st choice and the other half's last choice.

2. Uses More Information

In standard plurality voting, any information about a voter's 2nd, 3rd, or 4th choice is discarded. The Borda Count captures the full spectrum of a voter's preferences.

Why it matters: It provides a more accurate reflection of the "collective will." If a voter likes Candidate A and Candidate B almost equally, the Borda Count allows them to support both (to different degrees) rather than forcing them to choose only one.

3. High Social Utility

In mathematical simulations of voting behavior, the Borda Count often ranks very high in "Social Utility Efficiency." This means the winner is usually the person who provides the highest "average satisfaction" across the entire group.

4. Mathematical Monotonicity

The Borda Count is monotonic, which is a property that some other popular systems (like Instant Run-off Voting/Ranked Choice Voting) lack.

Why it matters: Monotonicity means that if a candidate's ranking improves on some ballots (and stays the same on others), their chance of winning cannot decrease. In some other systems, it is mathematically possible for a candidate to lose because they gained more support, which is often seen as a failure of fairness.

5. Reduces the "Spoiler Effect"

Because voters can rank all candidates, the presence of a "spoiler" (a third-party candidate similar to a major candidate) is less likely to hand the election to an opponent. Instead of splitting the vote, the similar candidates will likely be ranked 1st and 2nd by the same voters, allowing both to accumulate points.

6. Simplicity of Calculation

Compared to more complex systems like Condorcet methods (which require pairwise comparisons) or Instant Run-off (which requires multiple rounds of elimination), the Borda Count is very easy to explain and calculate. You simply add up the scores and the highest number wins.

7. Proven Success in Specific Contexts

The Borda Count is highly effective in environments where consensus is more important than "majority rule," such as:

Sports Awards: It is used to determine the MLB Most Valuable Player (MVP) and the Heisman Trophy winner.

Academic and Small Group Settings: It is frequently used by hiring committees or clubs to find a candidate that the entire group can get behind.

International Politics: Countries like Nauru and Kiribati use variations of the Borda Count for their national elections.

Note on Strategy: While these are the advantages, the Borda Count is sensitive to "strategic voting" (where voters rank a strong rival last to hurt their score). Because of this, it is generally considered best for groups where voters are likely to be honest or in contexts like sports where "voter intent" is public.